

Syllabus – MGTP 495 Accounting Data Analytics

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Rady School of Management

Note: The syllabus may be subject to minor changes regarding its outline.

Course Information

Course Number	MGTP 495
Course Title	Accounting Data Analytics
Quarter	Spring 2026
Class meetings	Monday, 1:00pm to 3:50pm
Classroom	Otterson Hall 1E106

Contact Information

Professor	Mario Milone
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Email Address	mmilone@ucsd.edu
Teaching Assistant	TBD ()
Office Hours	TBD

Textbooks and Readings

Textbook 1 (ISL in Python)	<i>An Introduction to Statistical Learning with Applications in Python</i> by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani Link
Textbook 2 (ISL in R)	<i>An Introduction to Statistical Learning with Applications in R</i> by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani Link
Textbook 3 (SB)	<i>Reinforcement Learning: An Introduction</i> by Richard S. Sutton and Andrew G. Barto Link

Course Description

The goal of this course is to provide a basic understanding of recent technologies used in data analytics, machine learning, and more broadly artificial intelligence. The course covers a wide range of statistical and artificial intelligence tools, from regression analysis to machine learning as well as reinforcement learning, with an emphasis on the use of these tools in the accounting profession and on accounting data.

The techniques and vocabulary developed in this course are useful for:

- Understanding high level concepts of machine learning and predictive analytics.
- Developing an intuition of how ML can be used in practice and its limitations.
- Interacting with data science engineers.

Programming language

The course will be taught primarily in the python language, but students are welcome to use their language of choice for the assignments and exams.

Course Objectives

At the close of this course, you will be able to:

- Run simple regression analysis.
- Understand the main concepts of machine learning methods.
- Setup and train various machine learning algorithms.
- Understand the fundamentals of reinforcement learning.
- Understand how neural networks integrate with machine learning and reinforcement learning.

Course Preparation and Requirements

This course requires weekly preparatory readings and regular group assignments. The assignments are necessary to properly digest the concepts presented in class, and the outline below precises which concepts are used for each assignment. The readings and assignments of a given session is supposed to be done and submitted before the start of the session. For instance, **students are required to read chapter 1 of the ISL book before the course starts.**

Course Evaluation

Evaluation in this course will be based on group assignments and a final exam. Additionally, students are expected to attend classes. The final grade will be determined as follows:

1 Attendance	See below
2 Group assignments	50%
3 Final exam	50%

1 Attendance

Attendance is required for this course. Students are expected to be on time and attendance will be recorded at the beginning of each session. Attendance will determine the maximum final letter grade that can be achieved, regardless of the grades received for participation, group case studies, or final case study. The table below explains the correspondance between the number of attended classes and the maximum letter grade. For example, if a student is present for 7 classes, she/he cannot be awarded a letter grade higher than B+ even if she/he receives full grades for participation, group case studies, and the final exam. Letter grades are first awarded based on the grading of participation, assignments, and final exam according to the table above, and will be capped by the student's attendance. The maximum letter grade of A+ will only be awarded to students that attend at least 8 classes out of 10 (if the grades received for participation, assignments and final exam are high enough).

Number of attended classes	Maximum Letter Grade
8 or more	A+
Between 6 and 8	B+
Between 4 and 6	C+
Between 2 and 4	D+
Less than 2	F

2 Group Assignments

Students are required to regularly prepare and submit written assignments (cf outline). The assignments with questions will be posted on Canvas at least a week before its due date. These questions are designed to help students understand the course material. Assignments can be done by groups of maximum 4 students. The assignments should be kept as brief as possible. Assignments should be uploaded to Canvas *before* the beginning of the class. They will be graded with a numerical grade between 0 and 100. **Note that any assignment that is not submitted will be counted with the lowest grade of 0.**

3 Final Exam

The final exam will be in the form of

- A multiple choice quizz.
- An assignment that students will have to do individually.

More information will be provided as we approach the end of quarter.

Dynamics of the syllabus

In order to promote the best learnings experience, this syllabus is subject to changes as the course progresses. Any change will be made in the interest of the students.

Course Outline

Part I – Regression Analysis

Session 1 – Introduction and Overview

March 30

Reading ISL Chapter 1

Assignment No assignment this week
There is no assignment for this class.

- Part I: Overview of data analytics and AI
- Part II: Introduction to Python

Session 2 – Linear Regression

April 6

Reading ISL 2.1.1-2.1.3, 2.2.1 and 3.1-3.3

Assignment Ungraded Lab (Topic: *Python Coding*)

Session 3 – Logistic Regression

April 13

Reading ISL 4.1-4.3

Assignment Assignment 1 (Topic: *Data Manipulation*)

Part II – Machine Learning

Session 4 – Bias-Variance Trade-Off and Regularization

April 20

Reading ISL 2.2.2, 6.1-6.2

Assignment Assignment 2 (Topic: *Linear Regression*)

Session 5 – CART and Random Forest

April 27

Reading ISL 8.1-8.2

Assignment Assignment 3 (Topic: *Logistic Regression*)

Session 6 – Neural Networks I

May 4

Reading ISL 10.1-10.2, 10.7

Assignment No Assignment

Session 7 – Neural Networks II

May 11

Reading ISL 10.1-10.2, 10.7

Assignment Assignment 4 (Topic: *CART and Random Forest*)

Part III – Reinforcement Learning

Session 8 – Dynamic Programming and Policy Gradient

May 18

Reading SB chapter 1 and 4

Assignment Assignment 5 (Topic: *Neural Networks*)

Session 9 – No class (Memorial Day)

May 25

Reading SB 9.3 and chapter 13

Assignment Assignment 6 (Topic: *Dynamic Programming*)

Session 10 – Conclusion

June 1

Reading SB 17.1

Assignment Assignment 7 (Topic: *Policy Gradient*)

Academic Integrity

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at: <http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

Upon any suspicion of cheating, an allegation report will be submitted to the academic integrity (AI) office.

You can find out more about the process at: <https://academicintegrity.ucsd.edu/process/>

Students with disabilities

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. No accommodations can be implemented retroactively.

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.